MANIPAL UNIVERSITY JAIPUR
(University under Section 2(f) of the UGC Act)

MANIPAL UNIVERSITY JAIPUR

Department **Information Technology**

Course Hand-out

A. Basic Details:

Programme Name:	B. Tech., Information Technology
Course Name:	Python Programming
Course Code:	INT2121
LTPC *(Lecture Tutorial Practical Credits):*	3 1 0 4
Session:	July-November 2025
Class:	III Semester
Course Coordinator:	Dr. Ganpat Singh Chauhan
Course Instructor(s):	Dr. Shikha Chaudhary, Ms. Nandani Sharma
Additional Practitioner(s) – if any *(Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.):*	To be identified and appointed later/ or at the beginning of the semester

B. Introduction: To build competence/aptitude of the student group familiarity with the course, and be able to use Python Programming to a variety of tasks

C. Course Outcomes: At the end of the course, students will be able to

CO Statement	*CO*	*Level*	*Target Attainment %*	*Target Attainment level*
[INT 2121.1]	Demonstrate the concepts of Python basics and control structures to write simple python programs.	Understand (L2)	75%	2
[INT 2121.2]	Implement Python programs using Python's data structures such as lists, tuples, sets, and dictionaries as well as creating user-defined functions.	Apply (L3)	70%	2
[INT 2121.3]	Perform file operations and error handling to develop robust programs.	Apply (L3)	80%	3
[INT 2121.4]	Utilize Object-Oriented Programming Concepts to create structured and modular code.	Apply (L3)	75%	2
[INT 2121.5]	Analyze and manipulate data using Numpy and Pandas.	Analyze (L4)	80%	3

Here, the level indicates the cognitive level of Revised Bloom's Taxonomy.

Information about attainment levels:

Attainment (%)	Level
< 60 %	0
≥ 60% < 70%	1
≥ 70% < 80%	2
≥ 80	3

D. Program Outcomes and Program Specific Outcomes:

[PO.1].	**Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
[PO.2].	**Problem analysis**: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
[PO.3].	**Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
[PO.4].	**Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
[PO.5].	**Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
[PO.6].	**The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.
[PO.7].	**Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
[PO.8].	**Ethics**: Apply ethical principles and commit to professional ethics_and responsibilities and norms of the engineering practices.
[PO.9].	**Individual and teamwork**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
[PO.10].	**Communication**: Communicate effectively_on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
[PO.11].	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
[PO.12].	**Life-long learning**: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

[PSO.1].	To apply innovative and creative techniques to design, simulate, implement complex IT solutions while leveraging existing and cutting-edge technologies.
[PSO.2].	To succeed to achieve inspiring IT oriented jobs and competitive examinations that offer promising & satisfying careers.
[PSO.3].	To recognize the importance of professional developments by pursuing postgraduate studies and positions.

E. Assessment Plan: Clearly write the criteria, its description, and associated marks for assessment of student achievements. In addition, the attendance requirements, assignments need to be mentioned in this section.

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	• 4 Quizzes • 2 Assignments	**30** (20+10)
End Term Exam (Summative)	**End Term Exam (Closed Book)**	**40**
	Total	**100**

F. Syllabus: Introduction to Python: Python Basics: Print statement, Comments, Variables, Keywords, Identifiers, Literals, operators • String Operations in Python • Simple Input & Output • Output Formatting Python Program Flow • Conditional statement • Looping Construct • Range Function• Break &Continue • Assert • Lists • Tuples • Sets • Dictionaries • Collections • Operations on Data Structures. Functions& Modules: User Defined Functions, Functions Parameters, Variable Arguments, Scope of a Function, Function Documentations, •Lambda Function Map, Filter and Reduce • Creation of Module. • File Handling •File handling Modes • Reading Files • Writing& Appending to Files • Handling File Exceptions • The with statement. Object and Classes: Classes in Python, Principles of Object Orientation, Creating Classes • Instance Methods. Exceptions Handling: Errors, Exception handling with try, handling Multiple Exceptions, Writing your own Exception Module. • Numpy: Introduction to numpy, Creation of vectors and matrices, Matrix manipulation, functions of Numpy. • Pandas: Introduction to Pandas, Pandas data structures – Series and DataFrame, Data wrangling using pandas, loading a dataset into a dataframe, Selecting Columns from a dataframe, Selecting Rows from a dataframe, Adding new data in a dataframe, Deleting data from a dataframe.

G. Lecture Plan:

Lecture No.	Topics	Session Outcome	Corresponding CO	Mode of Delivery	Mode Of Assessing CO
1.	Introduction to the course and Course handout discussion	Introduction to the course and Course handout discussion	NA	Lecture based teaching-learning	NA
2.	Introduction to Python: definition, Print statement, writing your first python program, Variables, Keywords, Identifiers,	Introduction to Python: definition, Print statement, writing your first python program, Variables, Keywords, Identifiers,	INT 2121.1	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
3.	Literals, operators, Comments, indentation	Literals, operators, Comments, indentation	INT 2121.1	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
4.	Internal and external typecasting in python	Internal and external typecasting in python	INT2121.1	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
5.	String Operations in Python	String Operations in Python	INT 2121.1	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
6.	Taking input from user, Output Formatting Python Program Flow	Taking input from user, Output Formatting Python Program Flow	INT 2121.1	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
7.	Ex1: Calculator using python	Ex1: Calculator using python	INT 2121.1	Learning through problem-solving	Mid-Term, quiz-1, ETE
8.	Conditional statement: if, if else	Conditional statement: if, if else	INT 2121.1	Learning through problem-solving	Mid-Term, quiz-1, ETE
9.	if elif else, logical operators	if elif else, logical operators	INT 2121.1	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
10.	Looping Construct: For, Range Function	Looping Construct: For, Range Function	INT 2121.1	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
11.	Looping Construct: While	Looping Construct: While	INT 2121.1	Lecture based teaching-learning	Mid-Term, quiz-1, ETE

12.	Break & Continue, Assert	Break & Continue, Assert	INT 2121.1	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
13.	Lists: introduction, creation, manipulation	Lists: introduction, creation, manipulation	INT 2121.2	Peer teaching	Mid-Term, quiz-1, ETE
14.	List operations, slicing	List operations, slicing	INT 2121.2	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
15.	Tuples: accessing, updating, unpacking a tuple	Tuples: accessing, updating, unpacking a tuple	INT 2121.2	Learning through problem-solving	Mid-Term, quiz-1, ETE
16.	Sets	Sets	INT 2121.2	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
17.	Dictionaries, accessing items, changing and adding dictionary items	Dictionaries, accessing items, changing and adding dictionary items	INT 2121.2	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
18.	Removing items, copying a dictionary	Removing items, copying a dictionary	INT 2121.2	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
19.	Collections, Operations on Data Structures	Collections, Operations on Data Structures	INT 2121.2	Peer teaching	Mid-Term, quiz-1, ETE
20.	For loop: with list, with string, with dictionary	For loop: with list, with string, with dictionary	INT 2121.2	Lecture based teaching-learning	Mid-Term, quiz-1, ETE
21.	User Defined Functions, Syntax, Functions Parameters	User Defined Functions, Syntax, Functions Parameters	INT 2121.2	Lecture based teaching-learning	Mid-Term, quiz-2, ETE
22.	Variable Arguments, Scope of a Function, Function Documentations	Variable Arguments, Scope of a Function, Function Documentations	INT 2121.2	Lecture based teaching-learning	Mid-Term, quiz-2, ETE
23.	Lambda Function Map	Lambda Function Map	INT 2121.2	Individual learning/ self-study	Mid-Term, quiz-2, ETE
24.	Filter and Reduce, Creation of Module	Filter and Reduce, Creation of Module	INT 2121.2	Lecture based teaching-learning	Mid-Term, quiz-2, ETE

25.	File handling, and its Modes, Reading Files	File handling, and its Modes, Reading Files	INT 2121.3	Lecture based teaching-learning	Mid-Term, quiz-2, ETE
26.	Writing & Appending to Files	Writing & Appending to Files	INT 2121.3	Individual learning/ self-study	Mid-Term, quiz-2, ETE
27.	Handling File Exceptions, The with statement	Handling File Exceptions, The with statement	INT 2121.3	Lecture based teaching-learning	Mid-Term, quiz-2, ETE
28.	Errors, Exception handling with try.	Errors, Exception handling with try.	INT 2121.3	Lecture based teaching-learning	Mid-Term, quiz-2, ETE
29.	handling Multiple Exceptions	handling Multiple Exceptions	INT 2121.3	Lecture based teaching-learning	Mid-Term, quiz-2, ETE
30.	Writing your own Exception Module	Writing your own Exception Module	INT 2121.3	Individual learning/ self-study	Mid-Term, quiz-2, ETE
Mid Term Examination					
31.	Classes in Python	Classes in Python	INT 2121.4	Lecture based teaching-learning	Mid-Term, quiz-2, ETE
32.	Principles of Object Orientation, Creating Classes	Principles of Object Orientation, Creating Classes	INT 2121.4	Learning through problem-solving	Mid-Term, quiz-2, ETE
33.	Instance Methods	Instance Methods	INT 2121.4	Group- teaching and learning	Mid-Term, quiz-2, ETE
34.	Introduction to numpy,	Introduction to numpy,	INT 2121.5	Lecture based teaching-learning	quiz-3, ETE
35.	Creation of vectors	Creation of vectors	INT 2121.5	Lecture based teaching-learning	quiz-3, ETE
36.	Creation of matrices	Creation of matrices	INT 2121.5	Lecture based teaching-learning	quiz-3, ETE
37.	Matrix manipulation	Matrix manipulation	INT 2121.5	Lecture based teaching-learning	quiz-3, ETE
38.	functions of Numpy	functions of Numpy	INT 2121.5	Learning through problem-solving	quiz-3, ETE
39.	Introduction to Pandas	Introduction to Pandas	INT 2121.5	Lecture based teaching-learning	quiz-3, ETE

40.	Pandas data structures – Series	Pandas data structures – Series	INT 2121.5	Lecture based teaching-learning	quiz-3, ETE
41.	Pandas data structures – DataFrame	Pandas data structures – DataFrame	INT 2121.5	Lecture based teaching-learning	quiz-3, ETE
42.	Pandas data structures – DataFrame	Pandas data structures – DataFrame	INT 2121.5	Lecture based teaching-learning	quiz-3, ETE
43.	Loading a dataset into a dataframe	Loading a dataset into a dataframe	INT 2121.5	Group- teaching and learning	quiz-3, ETE
44.	Loading a dataset into a dataframe	Loading a dataset into a dataframe	INT 2121.5	Group- teaching and learning	quiz-3, ETE
45.	Selecting Columns from a dataframe	Selecting Columns from a dataframe	INT 2121.5	Peer teaching	quiz-3, ETE
46.	Selecting Rows from a dataframe	Selecting Rows from a dataframe	INT 2121.5	Peer teaching	quiz-3, ETE
47.	Adding new data in a dataframe	Adding new data in a dataframe	INT 2121.5	Peer teaching	quiz-3, ETE
48.	Deleting data from a dataframe.	Deleting data from a dataframe.	INT 2121.5	Lecture based teaching-learning	quiz-3, ETE
End Term Examination					

H. Course Articulation Matrix: Couse articulation indicates the mapping of CO statement(s) to some PO/ PSO statement(s). Here value 0 indicates no correlation, 1 as low correlation, 2 as moderate correlation, and 3 as substantial / strong correlation.

CO	**STATEMENT**	**Correlation with Program Outcomes and Program Specific Outcomes**														
		PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO 1**	**PSO 2**	**PSO 3**
[INT 2121.1]	Demonstrate the concepts of Python basics and control structures to write python programs.	2	2	2	2	2							1	1	1	1
[INT 2121.2]	Implement Python programs using Python's data structures such as lists, tuples, sets, and dictionaries as well as creating user-defined functions	2	2	2	2	2							1	1	1	1
[INT 2121.3]	Perform file operations and error handling to develop robust programs.	1	2	2	2	2							1	1	1	1
[INT 2121.4]	Utilize Object-Oriented Programming Concepts to create structured and modular code.	1	2	2	2	2							2	2	2	1
[INT 2121.5]	Analyze and manipulate data using Numpy and Pandas.	2	3	3	3	3							1	3	3	1